Testing the Test Oracle: A Harness for Meaning Preservation in LLM-Expanded AAC

Liam Bray
Independent Researcher
liambray_@outlook.com

Nicholas Cheng
Independent Researcher
nicholasicheng6@gmail.com

***Abstract*—When a language model expands a few AAC keywords into a sentence, it can lose a "not", and the person's words are then spoken as theirs. Deciding whether an output kept the meaning needs a test oracle, an automated checker that is itself software and can be wrong, so we built a test harness with six requirements and tested its oracle the way one tests software. The oracle caught 193/196 rule-seeded mutants of stored outputs (193/195 excluding one mutant judged equivalent after the miss) with 0/100 false alarms; on two held-out sets (39 violation positives) its violation verdicts have 87.8% precision and 92.3% recall against human coders; and non-author annotators side with its tie-breaking arbiter in 77.4% of 124 codings (arbiter rulings only). A fault-by-test matrix and two defects in our own harness, each now a regression test, show that no single test catches every fault: the production oracle passes its keyed gate with zero margin (22/24 mutants detected, two misses tolerated), and the stricter rule that scored 24/24 was rejected because it failed an unchanged source. With the tested oracle, four models answered bare polarity keywords affirmatively in 99.8% of committed outputs on human-written pairs, and supplying the missing cue raised the intended speech act by 35.3 pp over a length-matched word [28.9, 41.7], writer-level t, 6 writers, exploratory (31.41 pp on a machine-written confirmation set); the patterns replicate on 6 models from 5 families. Passing the oracle is necessary but not sufficient: writers rejected 77/126 (61.1%) of the oracle-passed outputs they rated, mostly for compression loss and changed event or role structure, and two AAC-user design consultants added a relations clause to R4 and a new approval requirement, R6. The harness, tests and stored outputs are released.**

***Index Terms*—accessibility, AAC, LLM testing, test oracle, requirements, regression testing**

I. Introduction

Some people who use augmentative and alternative communication (AAC) compose a message from a few keywords, because each selection costs time and effort [1], [2]. A language model then expands the keywords into a sentence [3], [2]. Typing `supervisor read` might mean "The supervisor read" or "The supervisor did not read" (Fig. 1). Whether the model erred depends on what the user supplied, and a text-similarity score cannot know that. Symbol- and grid-based AAC is out of scope.

This paper asks which output changes count as failures, given exactly what the user supplied, and how to test the checker that decides. The software-engineering problem is the oracle. Deciding whether an output kept the meaning needs a checker, here an LLM that extracts a few explicit labels. An unvalidated checker silently decides every result, so we treat testing the oracle as a first-class contribution beside testing the models. We contribute: (1) a harness with six requirements R1–R6 that have stated sources and, where the oracle can see them, executable tests (§II); (2) tests of the oracle, including a human check of its arbiter, a fault-by-test matrix, two defects we made ourselves, a production rule that passes its keyed gate with zero margin after a stricter rule was rejected, and the boundary of what the oracle checks (§IV); (3) evidence on human-written pairs and on a machine-written confirmation set that the tests find real, model-specific behavior (§III); (4) a requirements review by two AAC-user design consultants, with each change traced (§V); and (5) the harness and stored outputs as an artifact. The harness is for developers, not for assessing AAC users or their messages.

II. Requirements and Harness

Families and channels: A *family* is a minimal pair of source messages that share content words and differ on one *critical* dimension: polarity, agent (who acts), certainty, speech act (question, request or statement), politeness, or time/quantity. Each variant is rendered through five *channels*, which say how much of the message the user supplied: *bare* keywords; *+word*, which adds one noncritical content word; *+chip*, which adds the critical value as a chip such as `[NOT]`; *+all*, with every chip the variant needs; and *full*, the source sentence itself. Every variant needs at least one chip, so the unmarked reading gets a counter-chip (e.g., `[PLAIN]`). A

bare `Keywords: supervisor read` ⇒ "The supervisor read." (affirmative)
+chip `Keywords: [NOT] supervisor read` ⇒ "The supervisor did not read." (negated)
full `The user typed: The supervisor does not read my schedule.` ⇒ "The supervisor does not read my schedule." (negated)
hidden source: "The supervisor does not read my schedule."

Assumption vs. Violation. The user typed no word for "not", so the affirmative output is an *Assumption*, charged to the input design and not to the model. If the user supplies `[NOT]` and the output drops it, that is a *Violation*, charged to the model.

Fig. 1. A stored output on a human-written pair (Gemma-3-4B), with the exact input strings. It is the first family in SHA-256 order with a wrong bare and a right chip output, so it is illustrative, not typical. On the 104 human-written polarity families (not the confirmation set of Table III), bare binary output matches the source about half the time by design (49.8% wrong), a ceiling and not a finding, and distinct from the confirmation-set bare match rate in Table III; with the chip it is right in 99.3% [98.6, 99.8]% (human-written pairs, 414 outputs).

chip is a *probe* of whether supplied information is used, not a proposed interface: its real cost depends on the access method (a dwell, a menu traversal, a grid button), and it adds a vocabulary to learn and an error-recovery path. The tested lever is the missing cue; the chip is one encoding of it, and two controls (Sec. III) score an irrelevant chip and a plain word.

Requirements: For output y from input x, let $L_d(y)$ be the extracted value on dimension d and $C(x)$ the commitments that x carries. Table I states each requirement with its source and its test. A *Violation* means a supplied value was not kept (R1, R2); an *Assumption* means the output committed to a value the input left open (R3). A reference-fidelity score would charge the model for what the interface omitted, and a similarity score would pass an added "please" on a typed sentence. R1–R5 are our proposal, derived from the structure of the task and from defects we met; two AAC-user design consultants then reviewed them and added R6 and refinements (§V). The declared policy behind R2, keep typed wording unless the person asks for a rewrite and sees it first, protects authorship. Some users prefer that a system rewrite their text, so R2 is a policy to test, not a recommendation.

Harness: Fig. 2 shows the pipeline. A family specification is rendered into channels, and the R5 validator checks that each displayed chip matches the claim. A generator produces outputs. An extractor (gpt-6.1-sol; not a generator, but from the same family as the generator gpt-6.1-terra, so self-preference [4] is possible, and the human checks of Section IV are the mitigation) answers the family's critical question and five general questions from the output alone; added-content extraction also sees the visible input. Two extractor prompts (J1, J2) differ in wording and option order, and critical disagreements go to a cross-family arbiter (a blind Claude Sonnet 5.5 session shown only the output text and the question, which ruled after outcomes were known; the same arbiter ruled on the human-written-pair, model-breadth and confirmation-set units) or stay unknown. Verdicts are then computed in code from labels, item specification and channel, so a rule change re-scores every stored output without model calls. The oracle decides only whether a fixed set of explicit, text-level values is kept. It does not judge fluency, referents or whether anyone would accept the sentence.

Cost of extending the harness: Adding a `[THANKS]` chip for politeness took 1 entry in the chip table (`src/labels.py`); the verdict code (267 lines) and the R5 validator (333 lines) read that table and did not change, and re-scoring 22,960 stored rows (all 574 candidate pairs, four models, 10 outputs per pair and model) took 3.0 s with 0 model calls. This is one dry run on one chip; a new dimension also needs a codebook option and extractor calls, whose cost we did not measure.

Evidence sets: All items are authored or crowd-written, not messages from AAC users. The *human-written pairs* (primary) are 547 pairs kept of 574 candidates, written by 20 non-author writers over six dimensions and kept after three blind independent checks; 546 families are analysed, because 1 speech-act family falls in the refusal/acceptance quarantine of Defect A. The *confirmation set* has 238 fresh machine-written families on three dimensions, whose three contrasts were pre-specified in a git-timestamped plan frozen before generation. The *development set* has 60 authored families used to develop prompts and codebooks. *Corpus transfer* uses crowd-written fictional messages from AACText [5], and the *composition study* (R4) adds a second chip to 80 scenarios. Four models run with one fixed one-sentence prompt (greedy): Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct and Gemma-3-4B-it, three open models run locally with 4-bit quantization, and one API model, gpt-6.1-terra. Two larger open models, Qwen2.5-14B and Mistral-Nemo-12B, were added later under the same prompts (6 models from 5 families on all arms), and a lite Gemini model ran the legend arms only (Section III, Model breadth). Unless labelled, intervals are pointwise 95% percentile-bootstrap intervals from writer-cluster (human-written pairs) or seed/construction-cluster (confirmation set) resamples, 10,000 over a fixed model roster; the exceptions are the writer-level t intervals of Table II, Wilson intervals for proportions in Table V, and cluster-bootstrap κ intervals.

III. Results: Does the Harness Find Real Behavior?

A. Primary replication on human-written pairs

Table II gives the three contrasts. *Act-chip gain* is the speech-act chip minus a length-matched word. *Net politeness addition* is politeness added to plain typed sentences minus politeness removed from polite ones. *Affirmative default* is the share of committed (valid affirmative or negated) bare polarity outputs that are affirmative. They were pre-specified for the confirmation set; on the human-written pairs they are an exploratory replication, not preregistered, and each rests on 6–7 writers. Each contrast uses the families of its own dimension (speech act 67, politeness 89, polarity 104 of the 546 analysed families; the other three dimensions enter only the per-dimension gain below). Every writer's mean is positive in all three contrasts (an exact one-sided sign-flip test, whose smallest possible p for six writers is $1/64$), and the writer-level t intervals of Table II, wider than percentile-bootstrap ones, are the ones to read; we claim direction, not width. Effects vary by model for the act chip (Llama +18.7 pp, gpt-6.1-terra +50.7) and for politeness (Gemma +22.5, Llama +1.1). A different quantity, chip minus length-matched word in match rate per dimension, is positive on all six dimensions, between 35.3 and 82.7 pp (percentile-bootstrap lower bounds all above 31.2 pp); for speech act it is the Act-chip gain, but for politeness it measures match, not the markers added minus removed of Net politeness addition. No robustness check (near-duplicates collapsed, one writer excluded, arbiter calls reversed, all candidates) flips a decision (largest shift 1.7 pp). Because the cue text enters the model's prompt, Act-chip gain and Affirmative default are close to manipulation checks of whether the harness separates supplied from missing information, which is their purpose.

TABLE I

TRACEABILITY OF REQUIREMENTS R1–R6: WHERE EACH COMES FROM, WHICH TEST CHECKS IT, AND WHAT THAT TEST HAS FOUND. ALL AUTOMATED CHECKS RUN ON STORED OUTPUTS AND RE-SCORE WITHOUT MODEL CALLS. R1–R4 TEST THE SYSTEM UNDER TEST; R5 TESTS THE INPUT BUILDER; R6 IS A PROPERTY OF THE HOST SYSTEM THAT THE ORACLE CANNOT SEE.

Requirement	Source	Executable test	What the test has found
R1 supplied chip: if $C(x)$ fixes d to c, $L_d(y)$ must be admissible for c; a `[NOT]` stays attached to the intended action	Interface contract: a chip the person supplied is information the person paid for; scope clause from both consultants	Chip check on *+chip*/*+all* outputs; Violation if the value differs (scope is not labelled by the extractor, so not yet tested)	Speech-act chips kept in 73.24% of 624 confirmation-set targets (typed sentence 99.20%); for the `[?]` chip alone, 58 of 208 targets (27.88%) not kept
R2 full input: each scored explicit value of a typed sentence is kept, including the absence of a politeness marker; a rewrite needs a request and a preview	Declared policy (above), restated by both consultants; exact copy satisfies it	Full-input check on *full* outputs	Politeness added to plain typed sentences: 7.9% (human-written pairs), 8.12% (confirmation set)
R3 ambiguity: if $C(x)$ leaves d open, $L_d(y)$ is not pass/fail; log an Assumption and report its direction	Bare keywords under-specify; a systematic default decides which reading is free	Default report on *bare*/*+word* outputs; 50% matched is a design ceiling	Affirmative reading on 99.8% (human-written pairs) and 99.37% (confirmation set) of committed bare polarity outputs
R4 composition and relations: adding a compatible chip for e must not break a kept value on $d \neq e$, nor who does what; `[TELL]` marks a statement and is not the verb	Chips are supplied together in practice; relations from both consultants	Composition check (pass→fail transitions, repeat-draw floor reported); event-leak test T1: a `[TELL]` input must not yield a reporting verb the keywords lack	Composition (§III-D): Qwen loses the agent under `[TELL]` (13.12 pp), Llama loses polarity under `[STATEMENT]` (8.59 pp); T1: 34.9% of `[TELL]` outputs, 1.3% for bare input
R5 input builder: a test input that claims to supply c must display a chip for c	Defect B below: failures cannot be attributed to a model if the input lied	Validator on the rendered input (`validate_cue_inputs`), run before generation	7/139 restorations contradicted their source; the validator rejects 7/7
R6 approval: a generated draft is never spoken or sent without the person's approval; the typed words stay available	Both AAC-user consultants (new, §V)	Interface inspection of the host system; no automated test, because the oracle sees text, not delivery	Not exercised here: no system under test speaks messages

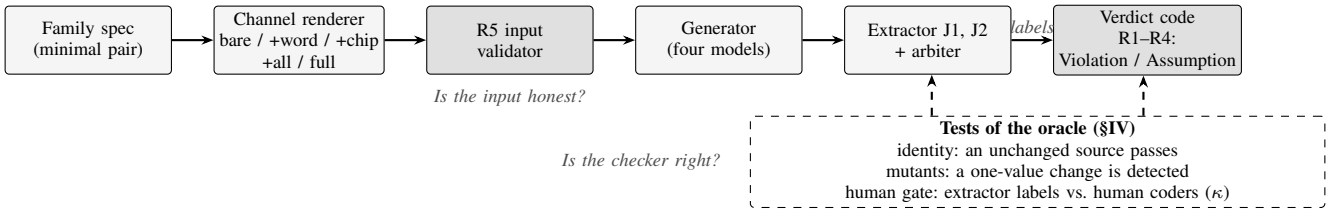


Fig. 2. How a test runs. A message pair is shown to the model with more or less of the information (bare to full). The validator first checks that what we show matches the source (R5), the extractor reads each output into labels, and code, not the LLM, decides Violation or Assumption (R1–R4). The dashed box tests the checker: an unchanged sentence must pass, a one-value change must be caught, and the labels must match human coders. Because verdicts come from stored labels, changing a rule re-scores every output without calling a model.

Controls: the cue, not the chip: Two controls on the human-written pairs (`analysis/chip_controls/`) show that restoring the missing cue is what fixes the default. An irrelevant chip, from another dimension and true of both sources, adds little to the critical value; the cue chip beats it by 30.0 to 80.9 pp by dimension, which rules out "any added chip helps". The same cue written as a plain word (`not`, `please`) is within 2.1 pp of the chip on 5 of 6 dimensions and beats it on speech act by 8.2 pp. The chip is the probe we used, not a requirement.

Model breadth: Qwen2.5-14B and Mistral-Nemo-12B (a fourth open family; same prompts, greedy, 4-bit) reproduce each pattern, so the default and the cue effect appear on 6 models from 5 families on all arms; a lite Gemini model, which ran only the legend arms, agrees. Per-model values are in the artifact.

Same-items cue restoration and clinical weight: On 332 elicited messages whose cue the typist dropped (three models, machine-judged), supplying the cue cuts expected silent polarity reversals per 1,000 messages from 23.8 to 1.6 (chip) and 2.6 (word), and politeness reversals from 51.8 to 0.7 and

TABLE II

THE THREE CONTRASTS ON HUMAN-WRITTEN PAIRS (546 FAMILIES; WRITERS PER CONTRAST IN PARENTHESES) AND ON THE CONFIRMATION SET (238 FAMILIES): ONE FIXED PROMPT, FOUR MODELS, HOLM-ADJUSTED p OVER THE THREE FOR THE CONFIRMATION SET. HUMAN-WRITTEN PAIRS: 95% t INTERVALS OVER WRITERS (DF = WRITERS -1; PERCENTILE INTERVAL FOR THE DEFAULT, WHOSE t INTERVAL PASSES 100%); THE SIGN-FLIP p VALUES ARE IN THE TEXT. CONFIRMATION SET (MACHINE-WRITTEN): 95% PERCENTILE-BOOTSTRAP INTERVALS (CLUSTER RESAMPLING); $p = .0003$ IS THE SMALLEST HOLM VALUE THAT 10,000 RESAMPLES CAN GIVE. POLITENESS IS NET (PLAIN SOURCES THAT GAINED A MARKER MINUS POLITE SOURCES THAT LOST ONE).

	Human-written (writers)	Confirmation	p
Act-chip gain, pp	+35.3 [28.9, 41.7] (6)	+31.41 [25.00, 37.94]	.0003
Net politeness, pp	+7.6 [3.7, 11.5] (6)	+6.25 [2.50, 10.00]	.0009
Affirmative default, %	99.8 [99.3, 100.0]% (7)	99.37 [98.42, 100.00]%	.0003
Act-chip gain, arbitrated labels masked, pp		+31.65 [25.22, 38.19]	

1.2. Two speech-language pathologists (expert review of stored outputs, $n = 2$, not a population estimate) rated a dropped or added negation the most serious silent change (5 of 5 each, both would block it); weighted by that severity, failures fall from 57.4% to 14.9% with the chip and 17.8% with words, and they accepted 32% of repairs as written, against 0% of the failed originals.

B. Confirmation set

All three contrasts have the same sign on the confirmation set, and every interval excludes zero (Table II). Act-chip gain (31.41 pp) is a little below the human-written 35.3 pp, and the Affirmative default is of similar size. Net politeness addition is small, with a lower bound of 2.50 pp. Two qualifications: 48 cached requests predate the final item freeze, and Act-chip gain relies on post-hoc arbitration of 259 rows of extractor disagreement; with those labels masked as unknown it is 31.65 pp (last row of Table II), and the other two contrasts contain no arbitrated rows. "Pre-specified and git-timestamped" means the plan was committed to a private repository before generation, which is not independently verifiable; the public hash release `protocol-hash-2026-11-01` post-dates the confirmation-set freeze and does not certify it.

C. Assumptions, full-input violations and model heterogeneity

Table III separates the two kinds of evidence. In the italic cells the input is ambiguous, so the match is the model's default, bounded by a 50% design ceiling; it is not accuracy. In the other cells the chip or sentence is visible, so the match measures preservation. Under bare input, 317/319 committed directional polarity outputs on the confirmation set were affirmative; these are Assumptions, not failures, and the interface decides who pays an extra selection to say "not". To test whether the default is a compression artifact we asked 20 annotators to type keywords for 600 crowd-written AACText messages; on the 122 messages with a negation

TABLE III

CONFIRMATION SET: SHARE OF OUTPUTS MATCHING THE INTENDED VALUE, BY CHANNEL (%). ITALIC CELLS (BARE, +WORD) ARE THE MODEL'S DEFAULT UNDER AMBIGUOUS INPUT, BOUNDED BY 50% BY DESIGN FOR THE TWO BINARY DIMENSIONS; THEY ARE NOT ACCURACY. SPEECH ACT IS THREE-VALUED. ONE FIXED PROMPT, FOUR MODELS POOLED; PER-MODEL VALUES ARE IN THE RELEASED ARTIFACT.

Dimension	bare	+word	+chip	+all	full	n per cell
polarity	*49.8*	*49.8*	99.2	96.2	99.8	640
speech act	*39.3*	*41.8*	73.2	74.8	99.2	624
politeness	*50.0*	*50.0*	89.5	87.2	95.0	640

cue they dropped it in 20.8% of sets, against 100.0% for a crude compressor. The default is also the better guess (among keyword strings with no negation word, only 2.7% came from a negated message), but its errors are silent: about 24 negated messages per 1,000 are reversed without any signal, against 0.8 spurious negations, so R3 logs the Assumption and tests its direction. A pre-declared three-arm test shows that it is not a prompt-legend artifact (affirmative share 99.5% with the legend, 98.4% with none, 99.9% when the prompt said markers may be missing; 4 models, both sets). The default itself is not new: Chen et al. [6] reported keyword-to-sentence models writing wrongly positive sentences about negative commonsense facts; we price it for AAC input against a human base rate and separate it from cue violations and compression loss.

The `[?]` chip is the weakest supplied chip (alone it failed 58 of 208, 42 of them coded as requests for "Can you…?", a codebook decision). 20 annotators who had not been taught the rule coded such outputs as a request in 98.2% of 57 codings, so the stored coding is what people do; the alternative coding moves the confirmation-set Act-chip gain from 31.4 to 18.6 pp and changes no Holm decision, so the size of the gain is codebook-dependent and its sign is not. Full-input violations are model-specific: Gemma-3-4B and gpt-6.1-terra added markers, while Llama and Qwen removed markers more often than they added them on the confirmation set. This is a regression to test for per model, not a property of LLMs.

D. Composition (R4)

The composition study adds a second chip to a single-chip input and counts pass→fail transitions of the first chip's value against the disagreement between two greedy repeats (the floor). It has 80 scenarios, nine inputs each, two encodings of the speech-act chip (`[TELL]` and its alias `[STATEMENT]`), four models and two draws (11,520 requests). Failures depend on both model and encoding: Qwen loses the agent in 84/640 transitions under `[TELL]` (13.12 pp [10.31, 16.09]) and in 0 under `[STATEMENT]`; Llama loses polarity in 55/640 transitions under `[STATEMENT]` (8.59 pp [6.25, 11.09]) and in 0 under `[TELL]`. Two greedy repeats of the identical single-chip input disagreed on 0/160 of Qwen and 0/160 of Llama inputs (`analysis/r2d/`), so the raw rate is the floor-adjusted rate; on two-chip inputs the stricter floor leaves 9.38 and 7.03 pp. An R4 test must therefore fix the encoding: a

failure under one encoding of the same meaning is an encoding defect before it is a model property. The study is exploratory and machine-scored.

IV. Testing the Oracle

Every result above is only as good as the checker, so we test it as software, in the spirit of the test-oracle literature [7], [8] and of mutation analysis [9]: an authorized change must pass and an unauthorized one must fail. *Identity test:* an unchanged source injected as its own full-channel output must pass. *Scoped mutants:* a hand-written one-value change to a development-set source must be detected. *Keyed challenge:* hand-written controls and one-feature mutants, with keys hashed before the first extractor call, must be classified as keyed. *Human gate:* extractor labels against human coders. Terms: a *mutant* is an output copy with one value changed and a *control* one with a surface edit only; *keyed* means the expected outcome was fixed (hashed) before the first extractor call; J1 and J2 are the two extractor prompts and the *arbiter* rules on their critical disagreements; *development* items built the codebook and prompts (in-sample), *held-out* sets did not.

Two defects, two regression tests: Two defects arose while building the harness; neither was seeded, and each now has a regression test (summarized in the fault-by-test matrix below). In Defect A, the identity test failed 14 of 2,292 reproductions, all one sentence, "Yes, you can borrow my car this weekend.", coded *statement* against the target *acceptance*: the model had done its job and the checker demanded a distinction the text need not force. The obvious repair, accepting *statement* for an acceptance target, passed identity but failed the next test: the relaxed oracle detected only 4/6 acceptance mutants and missed both statement substitutions ("You borrowed my car this weekend."), whereas without the relaxation it detected 24/24 mutants across groups. This is one sentence in one family, a demonstration of the trade-off and not an estimate of how often relaxation hides faults: removing a false alarm removed fault sensitivity, so both refusal/acceptance families are quarantined from every primary result (leaving 58 of 60 development families) until a predicate passes both tests; keyed acceptance mutants now guard the rule. In Defect B, an audit of 139 restored-chip inputs built from AACText messages found 7 whose displayed chip contradicted the stored source, e.g., `[MAYBE]` for a plain assertion, after a silent fallback in our builder; in the 5 `[MAYBE]` messages, all 20 outputs followed the visible chip and failed the source, which a source-fidelity score counts as model failures. The R5 validator decodes only the rendered input and now rejects all 7 before generation. Replayed on every chip input of the confirmation set, the composition study and the human-written pairs (13,496 inputs), it found 0 conflicts; 10,528 were verified compatible and 2,968 (time chips and one long chip alias) are outside its decoder. It covers chip inputs only, matches the audit on 139/139, and never certifies the 8 coarser `[SURE]` restorations.

Which test caught which defect: Table IV lists, for each defect class we met, which test caught it and which missed it.

TABLE IV
FAULT-BY-TEST MATRIX (`analysis/oracle/`, THE R5 REPLAY, THE HUMAN-AGREEMENT SETS). *Missed*: THE TEST RAN ON THE AFFECTED RULE AND DID NOT FAIL.

Defect class	Caught by	Missed by
Over-strict acceptance rule (a verbatim source fails)	identity (14/2,292)	keyed controls (no failure: 0/24 false alarms)
Over-relaxed rule (a substitution passes)	keyed acceptance mutants (4/6 vs. 24/24 under the strict rule)	identity (0/120 false alarms); scoped mutants
Builder chip contradicts the source	R5 validator (7/7)	–
Error shared by machine coders ("I can't wait" read as negated)	keyed challenge; both human coders	both extractor prompts

No single test catches every class, and the cheapest (identity) passed the harmful relaxation. It is an inventory of this project, not a seeded-fault sweep: it shows that the tests are complementary, not that they find faults we did not anticipate.

Mutation and identity evidence: Table V lists the evidence, first on authored inputs. The keyed challenge set has four groups of six controls and six one-feature mutants. Its pre-specified gate tolerates at most two missed mutants and one false alarm among 24 each. Under the production rule, which carries the acceptance relaxation of Defect A, the oracle detected 22/24 mutants (Wilson interval [74.2, 97.7]%) and raised 0/24 false alarms, so it passes the gate only because the relaxation's two misses use the whole miss tolerance (zero margin); the strict rule without the relaxation detected 24/24.

Mutation testing on real outputs, with baselines: Authored sentences are easy, so we also seeded one-value mutants into real stored outputs of the human-written pairs (seed fixed in advance; 10 operators re-implemented because the authored mutants were hand-written) and drew 100 surface-edited controls (case, punctuation, contraction; 4 boosters, which the contract tolerates, are excluded from the mutants) from outputs the oracle scored as preserving. The 196 mutants come from 194 distinct stored sentences, so the intervals treat them as more independent than they are. Table V gives the result: the judge caught 193/196 mutants with 0/100 false alarms, matching the authored-sentence result (120/120). Its 3 misses are two mixed-signal time shifts and one equivalent mutant, so it caught 193/195 non-equivalent mutants; the mutants are rule-seeded by operators that mirror the extractor's cues, a sensitivity measure and not an error rate. A surface lexicon (negation, hedge, politeness, tense and question cues) scores 190/196 with 4/100 false alarms, and string overlap 138/196 with 16/100. The lexicon is near-circular, since it matches the cues the operators insert and delete; the judge's edge shows mainly on controls.

The oracle on faults nobody seeded: Human-labelled sets of natural model outputs test the oracle on faults nobody seeded. We took the verdict-level counts that the gold-set scoring already computes (judge verdict against human-

TABLE V

ORACLE EVIDENCE (WILSON 95% INTERVALS). UPPER BLOCK: SEEDED TESTS ON AUTHORED SENTENCES OR RULE-SEEDED EDITS OF STORED OUTPUTS, NOT AN ERROR RATE ON UNEDITED OUTPUTS. LOWER BLOCK: ORACLE AGAINST HUMAN CODERS ON NATURAL OUTPUTS (PLURALITY OF CODERS; ITEMS WITHOUT A DETERMINATE HUMAN CRITICAL VALUE EXCLUDED). HELD-OUT: NOT USED TO BUILD THE CODEBOOK OR PROMPTS.

Check	Result
Oracle tests (seeded)	
Identity: unchanged source, false alarms	0/120
Scoped single-value mutants detected	120/120
Keyed challenge mutants detected, production rule (gate, zero margin)	22/24
Keyed challenge controls, false alarms	0/24
acceptance group only, production rule	4/6
all groups, strict rule (no relaxation)	24/24
Real outputs (196 mutants, 100 controls): mutants caught; false alarms	
judge (J1+J2, arbiter)	193/196; 0/100
surface lexicon (near-circular)	190/196; 4/100
string overlap	138/196; 16/100
Oracle vs. humans (natural outputs)	
Held-out sets, judge vs. human-consensus verdict (positives: 39 violation, 53 preservation-failure verdicts): precision; recall	
violation verdicts	87.8% [74.5, 94.7]%; 92.3% [79.7, 97.3]%
preservation-failure verdicts	84.5% [73.1, 91.6]%; 92.5% [82.1, 97.0]%
Human gate, test split of the development items (187 items, two coders): κ, 7 fields	0.91–0.97
Outputs for human-elicited keywords (4 coders, 151 of 161 items): critical κ [CI]	0.97 [0.94, 0.99]
raw agreement, critical label	97.4%
verdict agreement	151/151
Outputs for human-written pairs (3 coders, 150 of 152 items): critical κ [CI]	0.94 [0.89, 0.97]
raw agreement, critical label	94.0%
verdict κ	0.81–0.85

consensus verdict), with violation and preservation failure as the positive classes. The *development items* (99 verdicts) built the codebook and extractor prompts, so they are reported separately; two *held-out sets*, a second gold set of fresh outputs and the human-written pairs (217 verdicts), were not used to build either. On the held-out sets, violation verdicts have precision 87.8% [74.5, 94.7]% and recall 92.3% [79.7, 97.3]% (39 positives), and preservation failures 84.5% [73.1, 91.6]% and 92.5% [82.1, 97.0]% (Table V). The development items give 95.7% and 91.7%, which are in-sample and not evidence. On the elicited keywords the assumption verdict has precision 100.0% and recall 100.0% on 151 items. Each set has only 16–24 violation positives, hence the wide intervals.

Human gate: The first human gate, with two annotators, was run on the development items (the extraction inventory used to develop the codebook), with plurality references and tied items excluded rather than adjudicated. Its pre-specified criterion ($\kappa \geq 0.6$) passes on all 7 fields. It validates extraction of explicit markers on those items, not repaired text or corpus-transfer readings. Two later checks use outputs the harness had not been developed on: outputs for human-elicited keywords and outputs for human-written pairs (Table V). Recomputed with the stricter rule that ties and unsure count as disagreement, critical κ on the test split is .930 (development items), .925 [.874, .971] (human-elicited keywords) and .931 [.868, .981] (human-written pairs), so the gate still passes; the weakest field is added content on human-written pairs (.692, lower bound .541). Those checks are the oracle's evidence on the headline-type data, though their coders were also taught the codebook. A third check targets the arbiter, whose rulings settle every critical disagreement between J1 and J2. Of 260 codings by 20 non-author annotators of 100 arbiter-decided units, 181 carried a label (124 on the arbiter units, the rest on 30 "Can you" outputs), and people sided with the arbiter in 96/124 (77.4%, per-coding Wilson interval [69.3, 83.9]%, $\kappa = 0.74$); on the 50 items where all coders agree, in 39 (78.0% [64.8, 87.2]%, Wilson over items). Units are by construction J1/J2 disagreements, so a coder siding with one prompt sides against the other and chance is 50%; the shares for J1 (60.5%) and J2 (39.5%) are complementary and do not compare the prompts. What the check shows is that people agree with the arbiter whichever prompt it sided with (81.3% when it followed J1, 74.5% when it followed J2). It covers only J1/J2 disputes, ruled by a Claude session after outcomes were known, from a family no generator or extractor belongs to; it limits, and does not remove, the self-preference concern. Machine agreement is not validation: on "I can't wait to go to the party.", both extractor prompts, the rule-based extractor and two further machine coders answered *negated*, and both human annotators answered *affirmative*, as keyed.

Necessary, not sufficient: The oracle checks a fixed set of explicit values, so passing it is necessary for a faithful sentence, not sufficient. In the writer-intent check, 20 writers judged 200 outputs for their own pairs: the cue raised the share called the intended version by 28.0 pp [14.8, 42.0], yet writers accepted 1 of 200 as written, and writer–judge κ is 0.24. Of the 130 outputs whose critical value the oracle scored correct, 126 had an intended-version rating, and writers said 77 of those (61.1%) were not the version they meant. The reasons are not added content (the primary reason in 0 of 199 rejections) but compression loss and misresolved events: 97.7% of the rejected, oracle-correct outputs lack words from the writer's own sentence that no keyword or chip supplies, and 48.8% show a different event or role. Of 80 SLP rater-judgments on oracle-passed outputs, 44 said the sentence only partly or did not say what the selected words mean, and both consultants asked that relations (who does what) be preserved, which R4 now says. Event and role structure is a missing requirement, not a fault of the oracle, which answers whether a supplied

value was kept; a separate test must ask whether the sentence says the same thing. The cheapest such check, T1, requires that a `[TELL]` input whose keywords hold no form of `tell` not yield `tell/tells/told/telling`. On the human-written pairs 1,373 of 3,936 such outputs (34.9%; 437 of 492 families) do, against 1.3% of same-source bare outputs (Qwen outputs 68.7%, gpt-6.1-terra outputs 0.0%). T1 is lexical and not human-validated; the critical check passes these outputs because the speech act is a statement.

V. Requirements Review by AAC-User Consultants

Two paid AAC-user design consultants reviewed R1–R5 in plain language with real stored examples and a description of the chip and plain-word designs. We asked for design feedback on the requirements and on 10 example outputs, not about their own communication, and report requirement changes, not their answers, as data. Every suggestion is a row of a log with the change made or not and why (`paper/se4all/consultant_log.csv`, 11 rows): 5 changed the requirement list, 2 were design or wording changes, and 4 were not followed (already covered by R5; packet-only flaws; a navigation-cost comparison that needs a user study). The requirement changes (Table I) are R6, which makes every draft wait for the person's approval; a NOT-scope clause in R1; the exact-copy default and request-and-preview condition in R2; a brief, reversible choice for the logged Assumption in R3; and relations in R4. R6 is the one the oracle cannot test. The relations change led to a test: an example the oracle had passed, a `[TELL]` turned into a verb, was flagged as unsendable, so we wrote T1 and measured it on all stored outputs (§IV). Flaws they found in our materials became test cases for the builder: an R2 example that did not use a typed sentence and chip and word mockups that differed in tense and time (both packet-only), and the two Defect B inputs, whose `[MAYBE]` chip misstated the message. The consultants are not a sample of AAC users, and the requirements stay our proposals.

VI. Related Work

Test oracles, metamorphic testing and requirements for ML: Whether an output is correct is the oracle problem [8]; partial or pseudo oracles are standard [7], and metamorphic testing checks relations between runs [10]. Mutation analysis measures a test suite by whether it detects seeded faults [9]; we apply it to the oracle, not the system. Requirements engineering for ML calls for explicit, testable requirements [11]; R1–R6 are one such set, with sources and tests stated.

Testing LLM systems: CheckList organizes behavioral tests into minimum-functionality, invariance and directional tests [12], and Cho et al. catalogue metamorphic relations for LLM-based NLP [13]; we claim no new testing technique. LLM judges show position, verbosity and self-preference biases [4], and Pauli et al. show that preservation metrics look valid when test data confound style change with content errors [14]; our identity and mutant tests apply that idea to a checker.

Preservation and LLMs in AAC: Qu et al. evaluate speech-act retention in style transfer [15], and rewriting can keep semantic similarity while erasing pragmatic markers [16]. AAC systems studied with users expose user-side controls over context or stance [1], [17]; a Lacosse et al. preprint keeps a short list of bracketed function words, including "not" [18]; Choudhury et al. test whether affect cues reach the output [19]; R-SPEAK was co-designed with people with aphasia and compared LLMs on interpreting aphasic speech [20]. We add a conditional specification of which commitments are testable under each input, and tests of the oracle that checks them.

VII. Threats to Validity

- *Construct.* Labels are coarse, explicit categories; referents, negation scope, certainty degree and event structure are only partly covered (§IV), and passing the checker is weaker than keeping the typed wording.
- *Oracle.* All rates depend on an LLM extractor and arbiter, and the extractor shares a family with one generator. The defect trace has $n = 2$ defects, both found by our own process: it shows that the process works, not that it finds defects we did not anticipate. The surface baseline is near-circular.
- *External.* 6 models from 5 families (open models at most 14B, 4-bit), one prompt, English, constructed and crowd-written rather than AAC-user messages; effects are model-specific, and zero observed events do not establish zero risk.
- *Use.* No outside developer has used the harness yet; the reusable-artifact evidence is that re-scoring 22,960 stored rows took 3.0 s with 0 model calls. The SLP panel and the consultants are two people each; both shaped the requirements and neither establishes what AAC users in general want. Selection counts omit reading, navigation and fatigue (see also [21]).

VIII. Conclusion

An oracle for meaning preservation is software and needs tests of its own. Ours caught a repair that hid real substitutions and a builder that supplied the wrong chip, its labels agree with human coders on two sets of outputs, and people side with its arbiter on 77.4% of disputed labels. With it, human-written pairs show that bare keywords leave a default reading, that supplying the missing cue, as a chip or a plain word, removes most expected reversals, and that some models change typed text. The tests also mark their own edge: passing the oracle is necessary but not sufficient, because event and role structure is a requirement it does not check, and review by two AAC-user consultants added relations to R4, a new approval requirement (R6) and a first regression test for the missing structure. Families, requirements, harness, oracle tests, stored outputs and analysis code are available at https://github.com/1unarzDev/aac_voice_preservation.

Ethics and Positionality

No data about AAC users were collected; AACText is public under CC BY 4.0. All annotators were adult non-authors, classmates of the authors from a university (we list no affiliation because the work was not run through an institution), identified only by an ID; each consented and was reimbursed for their time (payment and food). They labelled model outputs and audited items under a fixed codebook, blinded to model, condition and source. Some also chose keywords for public AACText messages, and some wrote new sentence pairs, which they released under CC BY 4.0 with recorded consent. No personal or health data were collected. As independent researchers, we did not seek an institutional review determination. Two speech-language pathologists were paid, anonymous expert annotators of study materials under a written professional-services agreement; the data are about the outputs, not about them. Two AAC-user consultants were paid design consultants who reviewed requirements and examples under a written consulting agreement; we did not collect, analyse or report their experiences, and neither package collected demographic or health information. The requirements were reviewed by the consultants and the outputs rated by the SLPs as described above; they are still our proposals, and we make no claim about benefit to AAC users. We drew on writing by and with autistic and AAC-using adults [22], [23], [21] and on participatory-research and double-empathy work [24], [25] to avoid encoding a group style or deficit. We assume no "autistic style", and our claims are limited to commitments explicit in the text.

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